



**Joint AEPTX, CPS, Oncor and CenterPoint
Texas 765-kV STEP: Eastern Backbone
Project – ERCOT Independent Review
Status Update**

Ben Richardson

RPG Meeting
November 11, 2025

Introduction

- AEP Texas (AEPTX), CPS Energy (CPS), Oncor Electric Delivery and CenterPoint Energy Houston Electric, LLC (CEHE) submitted the Texas 765-kV-STEP Eastern Backbone Project for Regional Planning Group (RPG) review in July 2025
 - This Tier 1 project is estimated to cost \$9.384 billion and will require a Certificate of Convenience and Necessity (CCN) filing
 - Estimated in-service date (ISD) is between 2030 and 2032
 - This project was identified and included in 2024 Regional Transmission Plan (RTP) in Bexar, Dallas, Bell, Kleberg, Rusk, Wharton, other counties in the North Central, South, Central, Southern, Coast, and East Weather Zones
 - This project aligns with the 2024 RTP 765-kV Strategic Transmission Expansion Plan (STEP) Core Plan
- CenterPoint Energy provided an overview presentation and ERCOT provided the study scope at the September RPG Meeting
 - <https://www.ercot.com/calendar/09252025-RPG-Meeting>
- ERCOT provided the status update at the October RPG Meeting
 - <https://www.ercot.com/calendar/10282025-RPG-Meeting>
- This project is currently under ERCOT Independent Review (EIR)

Project Proposed by AEPTX, CPS, Oncor and CEHE

- Construct a new Blu Lacy 765/345-kV Substation and Install;
 - Five (5) 765-kV Circuit Breakers in a ring bus configuration expandable to breaker-and-a-half configuration;
 - Four (4) 765-kV, 300 MVA Switch Shunt Reactors (100 MVA/phase);
 - Two (2) Bus Switch Shunts each with a Circuit Breaker; and
 - Two (2) Line Switch Shunts each with a Circuit Breaker
 - Eleven (11) 345-kV Circuit Breakers in a seven-position (7) breaker-and-a-half configuration; and
 - Three (3) 765/345-kV transformers rated to at least 2,403 MVA Normal and 2,772 MVA Emergency;
- Construct a new Watermill 765-kV Switchyard and Install;
 - Twelve (12) Circuit Breakers in a double-bus, double-breaker configuration; and
 - Unspecified reactive device on 765-kV line

Project Proposed by AEPTX, CPS, Oncor and CEHE (Cont.)

- At Existing Watermill 345-kV Substation:
 - Install six (6) Circuit Breakers in a breaker-and-a-half configuration
- At Watermill 765/345-kV Substation Install:
 - Three (3) 765/345-kV transformers, rated to at least 2,403 MVA Normal and 2,772 MVA Emergency
- Construct a new Martin Lake 765-kV Switchyard and Install:
 - Eight (8) Circuit Breakers in a double-bus, double-breaker configuration
 - Unspecified reactive device on 765-kV line
- At Existing Martin Lake 345-kV Substation:
 - Install four (4) Circuit Breakers in a double-bus, double-breaker configuration.
- At Martin Lake 765/345-kV Substation Install:
 - Two (2) 765/345-kV transformers, rated to at least 2,403 MVA Normal and 2,772 MVA Emergency.

Project Proposed by AEPTX, CPS, Oncor and CEHE (Cont.)

- Construct a new Hillje 765-kV Substation and Install:
 - Install eight (8) Circuit Breakers in a breaker-and-a-half configuration
- At Existing Hillje 345-kV Substation:
 - Expand breaker-and-a-half configuration by adding two (2) new bays
 - Install five (5) Circuit Breakers in expanded breaker-and-a-half configuration
- At Hillje 765/345-kV Substation Install:
 - Three (3) 765/345-kV transformers, rated to at least 2,403 MVA Normal and 2,772 MVA Emergency
 - Two (2) 300 MVAr Line Shunt Reactors
 - Two (2) 300 MVAr Bus Shunt Reactors

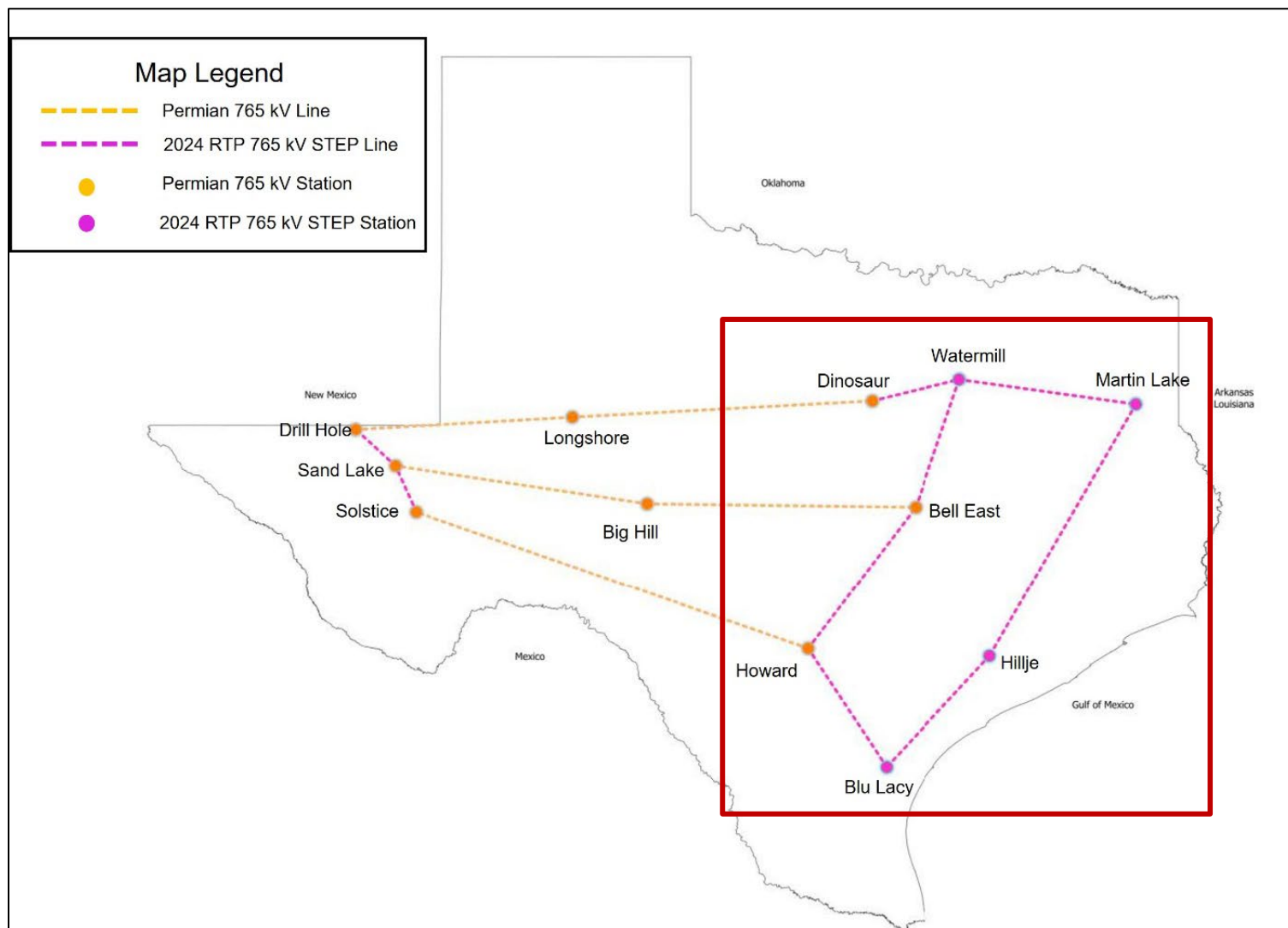
Project Proposed by AEPTX, CPS, Oncor and CEHE (Cont.)

- Install two (2) 765-kV Circuit Breakers in a double-bus, double-breaker configuration at New Substation 2 [Dinosaur]
- Install reactive device on 765-kV line at New Substation 2 [Dinosaur]
- Install four (4) 765-kV Circuit Breakers in a double-bus, double-breaker configuration at East Bell Substation
- Install reactive device on 765-kV line at East Bell Substation
- Install four (4) 765-kV Circuit Breakers in a double-bus, double-breaker configuration at Howard 765-kV Substation
 - Install two (2) 300 MVar 765-kV Line Shunt Reactors at Howard Substation
 - Install one (1) 300 MVar 765-kV Bus Shunt Reactor at Howard Substation

Project Proposed by AEPTX, CPS, Oncor and CEHE (Cont.)

- Construct a new Dinosaur to Watermill 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new right of way (ROW), approximately 78.0-mile;
- Construct a new Watermill to Martin Lake 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new ROW, approximately 152.4-mile;
- Construct a new Martin Lake to Hillje 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new ROW, approximately 292.8-mile;
- Construct a new Hillje to Blu Lacy 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new ROW, approximately 142.8-mile;
- Construct a new Blu Lacy to Howard 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new ROW, approximately 139.2-mile;
- Construct a new Howard to Bell East 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new ROW, approximately 170.4-mile; and
- Construct a new Bell East to Watermill 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new ROW, approximately 133.2-mile.

765-kV STEP Core Plan Proposed in 2024 RTP



Study Assumptions and Methodology

- The reliability need of this RPG project was identified and included in the 2024 RTP
- This RPG project aligns with the 2024 RTP 765-kV STEP Core Plan
- ERCOT will utilize the 2024 RTP study for this Independent Review and will assess the need for further analysis
 - Steady-state reliability analysis
 - Congestion Analysis
- ERCOT will perform additional evaluations in accordance with NERC TPL-001-5.1 and ERCOT Planning Criteria
 - Generation Addition and Load Scaling Sensitivity Analyses
 - Planning Guide Section 3.1.3(4)
 - Subsynchronous Resonance (SSR) Assessment
 - Nodal Protocol Section 3.22.1.3(2)

Reliability Analysis

- The 765-kV STEP Core Plan addressed the reliability issues identified in the 2024 RTP
- The 2024 RTP studies highlighted the benefits and cost effectiveness of the 765-kV STEP Core Plan
 - <https://www.ercot.com/calendar/01272025-EHV-765-kV-ERCOT>
 - Steady-state reliability studies
 - Significantly reduce power losses
 - Higher regional transfer capability
 - Sensitivity analysis with a reduced load level (~20 GW less overall load) showed major portions of the 765-kV STEP will still be needed to meet the reduced demand
 - Stability analysis
 - Increase the West Texas Export stability constraint limit
 - Improvement in system strength

Congestion Analysis

- Congestion analysis was also performed by using the 2024 RTP 2029 economic case for the 765-kV STEP Core Plan
 - 765-kV STEP Core Plan does not introduce significant new congestion in the study area
- The 2024 RTP studies showed that, in the long-term, 765-kV STEP Core Plan provides greater economic benefit than the traditional 345-kV plan
 - <https://www.ercot.com/calendar/01272025-EHV-765-kV-ERCOT>
 - Estimated in 2025 dollars, the TX 765-kV STEP reduces annual production cost by approximately \$28 million and produces approximately \$229 million more savings in consumer energy cost each year compared to the 345-kV plan in the long term

Generation Addition and Load Scaling Sensitivity Analyses

- Generation Addition Sensitivity Analysis
 - Per Planning Guide Section 3.1.3(4)(a), ERCOT performed a generation addition sensitivity by adding the new generation listed in Appendix A to the 2024 RTP 765-kV STEP Core Plan 2030 case. The additional resources were modeled following the 2024 RTP methodology. ERCOT determined relevant generators do not impact the AEPTX, CPS, Oncor and CenterPoint Texas 765-kV STEP: Eastern Backbone Project
- Load Scaling Sensitivity Analysis
 - Planning Guide Section 3.1.3(4)(b) requires an evaluation of the potential impact of load scaling on the criteria violations seen in this EIR. Starting 2024, ERCOT RTP adopted a new methodology of having one summer peak case for each study year with non-coincident peaks for each of the Weather Zones, which would eliminate the load scaling impact. The study case did not include load scaling as such load scaling sensitivity analysis is no longer needed

Subsynchronous Resonance (SSR) Assessment

- Subsynchronous Resonance (SSR) Assessment
 - Subsynchronous Resonance (SSR) Assessment was conducted for the AEPTX, CPS, Oncor and CenterPoint Texas 765-kV STEP: Eastern Backbone Project per Nodal Protocol Section 3.22.1.3
 - ERCOT found no adverse SSR impacts to the existing and planned generation resources at the time of this study

ERCOT Recommendation

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- At Hillje 765/345-kV Substation Install:
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 - Two (2) 300 MVAr Line Shunt Reactors
 - Two (2) 300 MVAr Bus Shunt Reactors

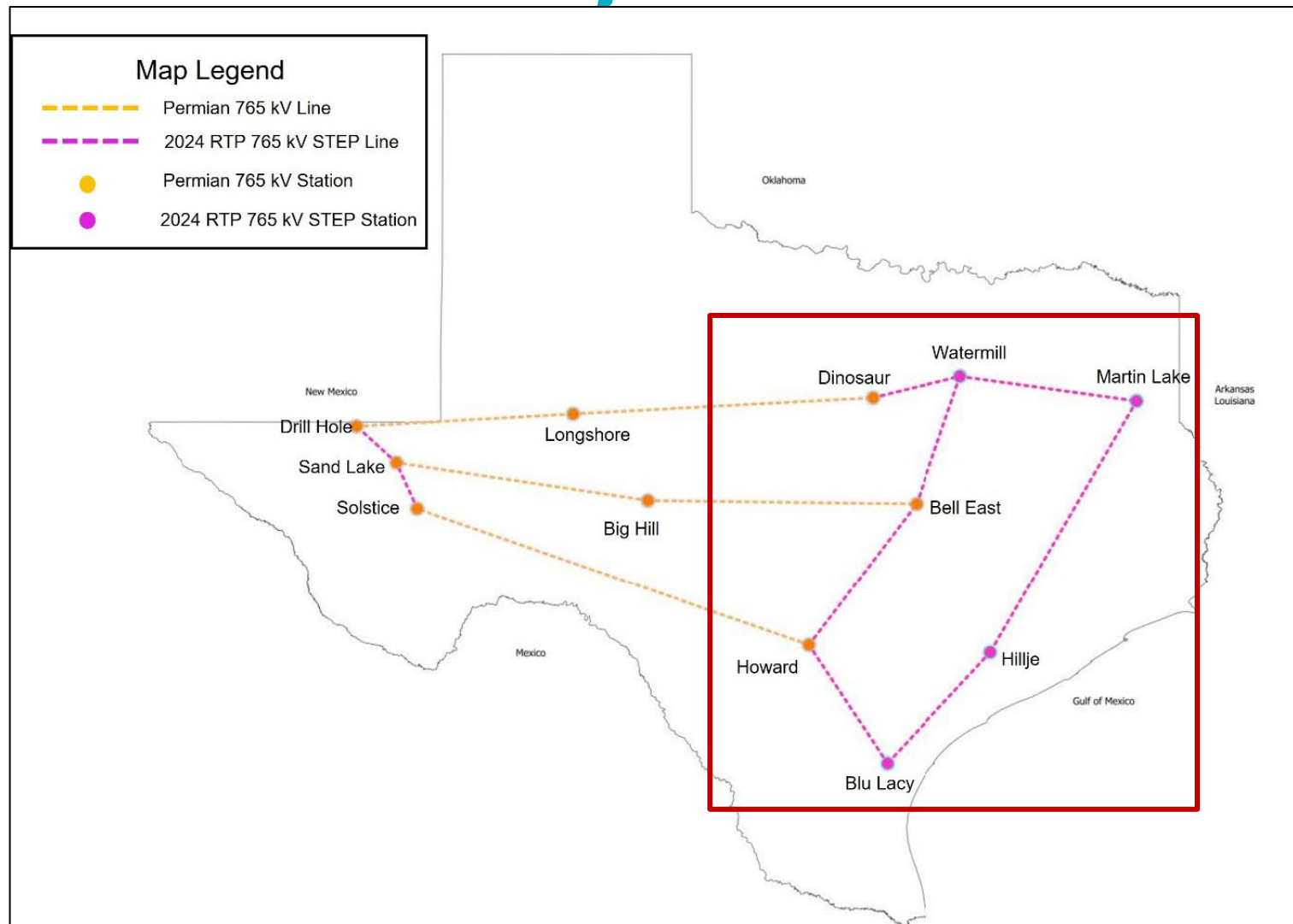
ERCOT Recommendation (Cont.)

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- Install reactive device on 765-kV line at New Substation 2 [Dinosaur]
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ERCOT Recommendation (Cont.)

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- Construct a new Martin Lake to Hillje 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new ROW, approximately 292.8-mile;
- Construct a new Hillje to Blu Lacy 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new ROW, approximately 142.8-mile;
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- Construct a new Howard to Bell East 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new ROW, approximately 170.4-mile; and
- Construct a new Bell East to Watermill 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new ROW, approximately 133.2-mile.

ERCOT Recommendation – Texas 765-kV-STEP Eastern Backbone Project



Next Step and Tentative Timeline

- Tentative timeline
 - EIR report to be posted in the MIS in November 2025
 - EIR recommendation to TAC in November 2025
 - Seek ERCOT Board of Directors endorsement in December 2025

Thank you!



Stakeholder comments also welcomed through:

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Appendix A – List of Units for Generation Addition Sensitivity Analysis

GINR	Project Name	Fuel	Projected COD	Max Capacity (~MW)	County
26INR0683	Z0 BECK ROAD DGR	OIL	09/09/2025	60.0	Bexar
26INR0675	A4 Pearsall DGR	OIL	09/10/2025	58.0	Bexar
26INR0694	V2 Brooks Field DGR	OIL	09/10/2025	76.5	Bexar
25INR0777	V4 Palo Alto DGR	OIL	09/11/2025	33.0	Bexar
26INR0681	K2 Nacogdoches DGR	OIL	09/17/2025	58.0	Bexar
26INR0676	Q1 Valley Road DGR	OIL	10/02/2025	29.0	Bexar
25INR0688	Giga Texas Data Center	OTH	12/15/2025	133.0	Travis
26INR0333	VERTUS ENERGY STORAGE	OTH	02/01/2026	207.3	Galveston
22INR0503	Tidwell Prairie II Batt	OTH	03/01/2026	203.6	Robertson
22INR0504	Barton Branch IA	OTH	03/01/2026	203.6	Robertson
24INR0533	Padua Grid BESS Unit 2	OTH	03/15/2026	150.9	Bexar
25INR0101	Mallard BESS	OTH	04/02/2026	257.3	Collin
25INR0706	Bocanova Power II	OTH	05/01/2026	150.5	Brazoria
28INR0024	Padua Grid BESS Unit 3	OTH	05/15/2026	201.4	Bexar
24INR0471	McGee Storage	OTH	05/29/2026	151.4	Henderson
26INR0405	Buffalo Creek BESS	OTH	06/01/2026	251.4	Fort Bend
24INR0584	Houston IV BESS	OTH	06/03/2026	164.6	Harris
25INR0166	Padrino Solar	SOL	07/30/2026	201.4	Hill
24INR0484	Lupinus Solar 3	SOL	09/21/2026	100.8	Franklin
24INR0490	Lupinus Storage 3	OTH	09/21/2026	50.9	Franklin

Appendix A – List of Units for Generation Addition Sensitivity Analysis (cont.)

GINR	Project Name	Fuel	Projected COD	Max Capacity (~MW)	County
26INR0375	Avalon BESS	OTH	12/01/2026	522.1	Fort Bend
26INR0474	Sorghum BESS	OTH	12/01/2026	208.9	Wharton
24INR0528	Blanquilla BESS	OTH	12/31/2026	200.8	Nueces
26INR0250	Duffy BESS	OTH	12/31/2026	235.9	Matagorda
25INR0170	Pinewood Solar	SOL	01/09/2027	150.7	Brazoria
25INR0253	Whiskey BESS	OTH	01/30/2027	153.7	Van Zandt
22INR0605	Camino Santiago Solar	SOL	02/18/2027	196.3	#SPILL!
23INR0266	Langer Storage	OTH	03/01/2027	61.5	Bosque
26INR0417	GAMAY ENERGY STORAGE	OTH	03/01/2027	418.6	Brazoria
24INR0188	Tehuacana Creek Solar	SOL	03/10/2027	505.4	Navarro
24INR0476	DOS RIOS ENERGY STORAGE SLF	OTH	03/15/2027	164.5	Milam
26INR0340	SOUTHERN SELECT ENERGY STORAGE	OTH	03/31/2027	404.3	Galveston
24INR0186	Rowdy Creek Solar	SOL	04/01/2027	351.8	Lamar
24INR0187	Rowdy Creek BESS	OTH	04/01/2027	175.9	Lamar
24INR0154	Lupinus Solar 2	SOL	04/03/2027	244.2	Franklin
24INR0155	Lupinus Storage 2	OTH	04/03/2027	122.9	Franklin
24INR0287	Rockridge Solar Project	SOL	04/03/2027	211.2	Guadalupe
25INR0164	Star Dairy Solar	SOL	04/03/2027	125.6	Lamar
25INR0334	Star Dairy Storage	OTH	04/03/2027	204.9	Lamar
25INR0382	HappyDogSolar	SOL	04/13/2027	85.5	Milam

Appendix A – List of Units for Generation Addition Sensitivity Analysis (cont.)

GINR	Project Name	Fuel	Projected COD	Max Capacity (~MW)	County
25INR0442	Happy Dog Storage	OTH	04/13/2027	104.5	Milam
24INR0285	APV Plumwood Storage	OTH	05/01/2027	200.8	Chambers
25INR0384	Zorro BESS	OTH	05/01/2027	153.7	Van Zandt
26INR0226	First Capitol BESS	OTH	05/01/2027	257.5	Brazoria
24INR0543	Thomas Cameron BESS	OTH	05/03/2027	201.5	Lampasas
25INR0304	Flying Kite Solar	SOL	05/21/2027	80.4	Zavala
25INR0391	Purple Sage BESS 1	OTH	05/30/2027	156.0	Collin
25INR0392	Purple Sage BESS 2	OTH	05/30/2027	156.0	Collin
25INR0235	Austin Bayou Storage I	OTH	06/01/2027	150.7	Brazoria
25INR0236	Austin Bayou Storage II	OTH	06/01/2027	156.6	Brazoria
25INR0237	Austin Bayou Storage III	OTH	06/01/2027	156.6	Brazoria
25INR0293	TOCE BESS 1	OTH	06/01/2027	205.2	Wharton
26INR0519	Sky Global West Houston	GAS	06/01/2027	324.6	Austin
27INR0321	Rock Island Generating (TEF -Due Diligence)	GAS	06/01/2027	121.8	Colorado
24INR0356	Cibeles Solar	SOL	06/30/2027	251.0	McLennan
25INR0215	Middlebrook BESS	OTH	06/30/2027	158.6	Nacogdoches
25INR0247	Bluebonnet Prairie Wind	WIN	07/15/2027	173.0	Navarro
25INR0319	Northington Solar	SOL	07/15/2027	129.8	Wharton
26INR0543	Three Canes Solar SLF	SOL	07/31/2027	333.0	Navarro
25INR0103	Elio BESS	OTH	08/05/2027	317.2	Brazoria

Appendix A – List of Units for Generation Addition Sensitivity Analysis (cont.)

GINR	Project Name	Fuel	Projected COD	Max Capacity (~MW)	County
22INR0220	Lamkin Solar	SOL	08/08/2027	101.5	Comanche
24INR0418	Middlebrook Solar Limited Liability Company	SOL	08/13/2027	316.2	Nacogdoches
25INR0370	Willis Storage	OTH	08/13/2027	176.1	Rains
23INR0090	Iguala Solar	SOL	09/01/2027	250.0	Victoria
24INR0150	Lupinus Solar 1	SOL	09/13/2027	164.9	Franklin
27INR0173	Trenno BESS	OTH	09/13/2027	202.6	Johnson
25INR0122	Vial BESS	OTH	09/15/2027	135.2	Hill
26INR0551	Hastings Park BESS	OTH	09/15/2027	256.3	Brazoria
22INR0426	LUBIO SOLAR	SOL	09/17/2027	54.5	Kaufman
25INR0204	Claxton Solar	SOL	09/17/2027	150.6	Hopkins
25INR0205	Claxton Storage	OTH	09/17/2027	75.7	Hopkins
26INR0380	Pepper Solar Farm	SOL	09/20/2027	120.7	McLennan
26INR0409	Lucky 7 Solar	SOL	09/20/2027	100.8	Hopkins
27INR0399	Resaca Oasis Storage	OTH	09/20/2027	200.7	Cameron
26INR0330	Alamo City BESS	OTH	09/21/2027	128.3	Bexar
26INR0180	Cape Rich BESS	OTH	09/23/2027	150.5	Navarro
25INR0494	Willow Beach Wind	WIN	10/01/2027	202.3	Brazoria
25INR0554	Dan Kearney BESS	OTH	10/01/2027	200.8	Brazoria
25INR0369	Echols Creek Storage	OTH	10/13/2027	100.6	Lamar
26INR0241	Sol Marina Energy Center	SOL	10/29/2027	175.3	Ellis

Appendix A – List of Units for Generation Addition Sensitivity Analysis (cont.)

GINR	Project Name	Fuel	Projected COD	Max Capacity (~MW)	County
26INR0242	Sol Marina Energy Center BESS	OTH	10/29/2027	57.2	Ellis
25INR0480	Bell Creek BESS	OTH	10/30/2027	200.9	Brazoria
26INR0314	Brazos River BESS	OTH	10/30/2027	180.4	Brazoria
25INR0281	Cosper Solar	SOL	11/12/2027	148.2	Bell
23INR0342	Brizo BESS	OTH	12/01/2027	140.8	Victoria
23INR0561	Cascabel Wind 2	WIN	12/01/2027	197.8	Zapata
24INR0424	Cascabel Wind 1	WIN	12/01/2027	197.8	Zapata
24INR0505	Corralitos Wind 1	WIN	12/01/2027	196.1	Zapata
25INR0422	Waterloo Solar 1	SOL	12/01/2027	200.6	Bastrop
25INR0447	Waterloo BESS 1	OTH	12/01/2027	105.6	Bastrop
24INR0269	Bar Candela Solar SLF	SOL	12/31/2027	200.3	Freestone
24INR0422	Hollow Branch Creek Solar	SOL	12/31/2027	460.0	Leon
24INR0506	Corralitos Wind 2	WIN	12/31/2027	195.4	Zapata
25INR0441	Dios BESS	OTH	12/31/2027	201.1	Jackson
24INR0178	Cumulus Grid BESS	OTH	01/01/2028	210.7	Ellis
23INR0137	BRP Diran BESS	OTH	01/31/2028	100.6	Wharton
24INR0065	Keys Hollow Solar Phase II SLF	SOL	03/10/2028	204.1	Goliad
24INR0067	KEYS HOLLOW SOLAR SLF	SOL	03/10/2028	204.1	Goliad
23INR0200	Paradiso BESS	OTH	03/15/2028	100.9	Atascosa
26INR0407	Rock Creek BESS	OTH	03/29/2028	251.1	Kendall

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GINR	Project Name	Fuel	Projected COD	Max Capacity (~MW)	County
27INR0243	Brookview Road BESS	OTH	05/02/2028	203.2	Harris
26INR0147	Seven Springs Solar	SOL	05/16/2028	145.0	Lampasas
26INR0431	Big Rooter West Solar	SOL	05/31/2028	403.4	Robertson
23INR0261	Dodd Solar I	SOL	06/05/2028	201.8	Henderson
26INR0074	Anila Solar	SOL	06/30/2028	200.8	Wilson
26INR0077	Anila BESS	OTH	06/30/2028	209.0	Wilson
25INR0330	Clutch City Solar Phase II	SOL	07/01/2028	251.3	Brazoria
27INR0224	Leopard BESS	OTH	07/09/2028	264.2	Victoria
22INR0466	Pajarita BESS	OTH	07/19/2028	205.5	Cameron
22INR0443	Bowstring BESS	OTH	09/18/2028	155.0	San Patricio
24INR0330	TRAVIS KIMBRO BESS	OTH	09/18/2028	36.8	Travis
27INR0389	Sugaree Solar	SOL	09/21/2028	251.4	Maverick
27INR0465	Althea Storage	OTH	09/21/2028	251.3	Maverick
27INR0126	Corvus Solar	SOL	11/12/2028	201.5	Webb
27INR0118	Grapefruit Solar	SOL	12/31/2028	291.4	Hidalgo
24INR0108	Louisa Energy Storage	OTH	03/12/2029	102.8	Bexar
23INR0479	Taormina Storage	OTH	05/26/2029	231.9	Bexar
29INR0017	Big Rooter East Solar	SOL	05/31/2029	554.9	Robertson
24INR0160	Huisache Solar	SOL	06/30/2029	125.6	Wilson
24INR0376	Huisache BESS	OTH	06/30/2029	100.7	Wilson

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GINR	Project Name	Fuel	Projected COD	Max Capacity (~MW)	County
28INR0127	Half Moon Solar	SOL	07/01/2029	512.5	Starr
23INR0467	Lavender Storage Project	OTH	09/13/2029	231.9	Bexar
24INR0156	Moonstone Solar Project	SOL	03/24/2030	145.8	Wilson